

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	560290
Award Program	Tipping Points
Project Title	Modeling the Future of Food in Your Neighborhood
Grant Start Date	04/01/2018
Grant End Date	02/28/2022
Total Grant Budget	\$2,046,273
Total FFAR Budget	\$1,036,418
Matching Funder(s)	Michael & Susan Dell Foundation, National Dairy Council, Nationwide Children's Hospital, Sears-Swetland Family Foundation, Sisters of Charity Foundation of Cleveland, Saint Luke's Foundation, Stephen J. McHale Family Foundation, Case Western Reserve University School of Medicine, Albert Einstein College of Medicine, Ohio State University John Glenn College of Public Affairs, and contributions of the study's Core Modeling Team
Final Report Date	06/01/2022
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Year 1 – 4/1/2018 - 3/31/2019

Year 2 – 04/01/2019-03/31/2020

Year 3 – 04/01/2020-03/31/2021

Year 4 – 04/01/2021-02/28/2022

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country. CLEVELAND, OHIO, OH-011
- 1.2. How many new jobs were created because of this grant funding?
 - a) Year 1 – 7
 - b) Year 2 - 7
 - c) Year 3 – 5
 - d) Year 4 - 0
 - e) Total - 19
- 1.3. How many jobs were maintained because of this grant funding?
 - a) Year 1 – 13
 - b) Year 2 - 12
 - c) Year 3 – 17
 - d) Year 4 - 13
 - e) Total - 55
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain. N/A
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization. N/A

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Case Western Reserve University was generously funded by Foundation for Food & Agriculture Research (FFAR) and matching partners from April 2018 to February 2022 to conduct the Modeling the Future of Food in Your Neighborhood study (foodNEST 2.0). The study was funded under the FFAR "Tipping Points Program" designed to support research that identifies leverage or tipping points in food systems where specific changes can improve overall community health and the economy. The goal of our research was to create complex systems models that reflect interdependencies among different actors and interventions influencing local food system dynamics and health equity in racialized urban neighborhoods. The study took place in Cleveland, Ohio building on the city's rich context of food systems innovation and strong community-engaged research and data systems infrastructure of the university. Collaborators included researchers from Ohio State University and Albert Einstein College of Medicine as well as a strong and diverse team of more than two dozen community partners.

The foodNEST 2.0 study applied a participatory system dynamic modeling approach to examine how changes within the local food system interact to influence health equity. The primary goal was to unpack the elusiveness of food systems change and guide future efforts related to community and neighborhood health and development. To achieve this goal, the team sought to develop validated tools that are user-friendly, relevant, and action oriented. The research approach included methods to engage community partners in the development of models reflecting food system dynamics that started as qualitative depictions of relationships within the local food system and then moved to quantitative heuristic applications to evaluate different strategies over short and long-term timeframes and under different neighborhood conditions. This resulted in the identification of a new, aspirational goal to structure behavior and outcomes of food systems in racialized neighborhoods—nutrition equity. The community-academic research team defined nutrition equity as having freedom, agency, and dignity in food traditions resulting in people and communities healthy in body, mind, and spirit. The team translated its modeling findings into a user-friendly "Menu of Actions for Community Driven Food Systems Change" that documents places to intervene in the system to foster nutrition equity through synergistic actions across different sectors (e.g., economics, health, education). The team applied this tool by implementing a fellowship program to work in solidarity with local leaders as they developed strategies to increase nutrition equity locally and beyond. This resulted in the creation of the "Vision Board Toolkit for Food Systems Change" that provides a step-by-step process for creating a transformation proposition to realize community-driven food systems change through local actions. These key products – the concept of nutrition equity, Menu of Actions, and Vision Board Toolkit – are designed to accelerate health equity by promoting access to basic food needs with dignity, balancing supply and demand of fresh and healthy foods, and expanding community

ownership of local food systems in racialized neighborhoods. All concepts and tools developed through foodNEST 2.0 have the potential to bolster community health through equitable advancement of local food systems.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Findings from the Modeling the Future of Food in Your Neighborhood Study provide strategic guidance to the agriculture community's commitment to racial equity offering a response to redress entrenched inequities in food and nutrition-related outcomes. Specifically, this research provides insight for leveraging and linking the food and agriculture sector to advance community health and economic opportunity in racialized urban neighborhoods. Our systems modeling study identified nutrition equity as a new goal to guide local food systems; a finding salient during this pivotal point in time when the 50+ year anti-hunger strategy of the US is being re-visioned. Nutrition equity was an emergent concept of the participatory system dynamic modeling research process reflecting an aspirational state of having freedom, agency, and dignity in food traditions resulting in people and communities healthy in body, mind, and spirit. This new concept can be viewed in contrast to a goal of food security and existing policy (e.g., SNAP) and programmatic responses (e.g., nutrition incentives), which have failed to lift people out of vicious cycles of poverty and provide sustained nourishment to promote health and economic opportunity. Systems science suggests that changing the goal of the system represents the greatest opportunity for transformation. Our findings related to nutrition equity may guide transformative efforts at the intersection of food systems and health.

Our new concept of nutrition equity and its related feedback mechanisms resonate with paradigm shifts within anti-hunger work, especially global calls for transforming the food system as a whole. The emergent concept of nutrition equity adds to national and global anti-hunger revisioning efforts in four critical ways. First, it roots local food systems and anti-hunger efforts in values of freedom, agency, and dignity, which act as a countervailing force to five decades of policies and practices focused on basic food needs that are underpinned by a different set of values. Second, it identifies a three-dimensional target for food systems and anti-hunger efforts in racialized neighborhoods focused on economic opportunity, food security, and fair access to fresh and healthy foods. Anti-hunger efforts should be accountable to these three outcome measures. Third, it reinforces the fact that there is not one silver-bullet solution to complex problems like food and nutrition insecurity. Our nutrition equity system dynamics model illuminated multiple yet inter-connected mechanisms to target within anti-hunger initiatives, including those focused on (1) meeting basic food needs with dignity; (2) balancing supply and demand of affordable, fresh, and healthy foods in neighborhoods; and (3) promoting community empowerment and ownership of local food systems change. Synergistic investment in these mechanisms in a manner that is responsive to local assets and opportunities and engages community leadership to guide implementation has the potential to

accelerate nutrition equity. Fourth, it emphasizes healthy body, mind, and spirit as a holistic target for evaluating the impact of local food systems and anti-hunger initiatives.

Our findings related to nutrition equity offer guidance to the food and agriculture sector about processes (i.e., what will we do?) and outcomes (i.e., what will we achieve?) of anti-hunger efforts. From a process perspective, findings shed light on the need for local food systems and anti-hunger efforts that foster meaningful community engagement in all stages of decision making and implementation, with targeted engagement from those most disaffected by the food system as it is now (i.e., communities of color, people with low-income). Furthermore, it calls for cross-sector collaboration and synergistic efforts, such as initiatives that link job training with emergency food assistance or community and economic development with food entrepreneurship. From an outcomes perspective, achieving the goal of nutrition equity will require collective investment in and evaluation of the “three trifectas” of nutrition equity within targeted community settings, including the extent to which local food systems and anti-hunger initiatives: (1) result in economic opportunity, food security, and fair access to fresh and healthy foods within neighborhoods; (2) lead to healthy body, mind, and spirit within communities; and (3) facilitate freedom, agency, and dignity in food traditions through community-driven change. A commitment to nutrition equity as a North Star goal for the food and agriculture sector also will require outcome assessment of community engagement processes, such as who is involved in decision making related to local food systems and anti-hunger efforts, who is getting funded to do this work, and the extent to which these efforts center the lived experiences of people of color and those with low income.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The Modeling the Future of Food in Your Neighborhood Study sought to develop decision support tools that offer a systems lens within an equity framework to guide local food systems change. Key insights from the community-engaged system dynamic modeling study were translated into two toolkits to promote equitable improvements to local food and agriculture sectors. The toolkits were designed for use by diverse stakeholders, including community residents, food and nutrition practitioners, anti-hunger groups, local government, education systems, and philanthropy.

The first toolkit, the “Menu of Actions for Community Driven Food Systems Change,” describes a process for engaging community stakeholders in decision making designed to transform local food system dynamics in racialized urban neighborhoods. The tool emphasizes the use of collaborative processes ensuring representation of affected groups in decision making about local food systems work and seeking to build trust among collaborators. Moreover, it highlights the importance of ongoing examination of unintended consequences of food systems change and the need for adaptive responses that account for dynamics of the real world. The Menu of Actions identified five leverage points for tipping the food system to fairness for people and communities disproportionately burdened by the food system as it is now (i.e., people of color and low-income populations). These highlight the role of cross-sector collaboration to accelerate equitable outcomes related to economic opportunity, food security, and fair access to fresh and healthy foods with the broader goal of promoting holistic health within communities through the food and agriculture sector. More than 40 potential community-driven actions were identified related to the five leverage points, such as innovations to make institutional (e.g., school, hospital) meals healthier and more culturally relevant; expanding neighborhood-based nutrition equity response systems; developing and expanding online ordering, food delivery, and nutrition incentives for vulnerable populations; growing food for diverse markets and culinary endeavors; enhancing perishable food distribution and recovery; food justice training and education for children and youth; and coordination of policy change to sustain food systems initiatives.

The second toolkit, the “Food Systems Change Vision Board Toolkit,” provides a step-by-step process to engage community leaders in efforts to translate their visions for food systems change into community-driven action. The toolkit is informed by the Business Model Canvas, a widely used tool designed to support entrepreneurial ideation and innovative action planning. The Vision Board Toolkit revolves around the goal of nutrition equity, an emergent concept resulting from the participatory system dynamic modeling approach employed by the Modeling the Future of Food in Your Neighborhood Study. Accordingly, it facilitates community-driven food systems changes related to the three main dynamics of nutrition equity including: meeting basic food needs with dignity, balancing supplying and demand of fresh and healthy foods, and community empowerment and food sovereignty. Through an iterative process, use of the toolkit supports the development of a transformation proposition to shift dynamics of local food systems to advance freedom, agency, and dignity in food traditions resulting in holistic health for people and communities. In 2020, five local leaders in Cleveland, Ohio applied the Vision Board Toolkit to support ideation, partner formation, and action planning related to community-driven efforts, such as those promoting culinary curiosity among children within childcare centers, urban farming and community ownership of land, and community engagement of a volunteer base to ignite food systems efforts in racialized neighborhoods.

Another impact of the research was the development of the nutrition equity system dynamic model that was published in the *American Journal of Clinical Nutrition* (Freedman et al., 2022). This model highlights three domains of feedback and 10 feedback mechanisms structuring nutrition equity in racialized urban neighborhoods and offers guidance for anti-hunger and local food systems efforts seeking to tip the food system to fairness in racialized neighborhoods. The three domains of feedback are related to a series of actions that accelerate or delay accessing basic food needs with dignity, balancing supply and demand of fresh and healthy foods, and expanding community empowerment and ownership of local food systems. The domain that has the broadest potential for impact on nutrition equity is the community empowerment and food sovereignty domain due to its effect on infusing community participation in implementation and adaptation of anti-hunger and food systems work. Findings point to the need for implementation and evaluation of policies and resulting programmatic initiatives that seek to advance community ownership of local food systems to examine the extent to which these community-driven initiatives accelerate economic opportunity, food security, and fair access to fresh and healthy foods as well as community health and wellness.

An editorial was published in the *American Journal of Clinical Nutrition* based on our nutrition equity system dynamic model (Singleton, 2022). The editorial reinforces how the nutrition equity system dynamics model has the potential to shape urban food systems and policy actions seeking to redress inequities related to food security, nutrition, and diet-related health outcomes. The editorial also draws attention to the exogenous factors identified during the system dynamic modeling process including social, political, and environmental factor (i.e., rising cost of living, policing, homelessness, underperforming schools) that impede our ability to improve food systems in historically redlined communities. The editorial concludes by stating: "In summary, future interventions and policy action aimed at combating the historical effects of structural racism on urban food systems must make nutrition equity the primary objective. This step is necessary to improve the diets and health of people of color in America who live in communities that continue to experience the manifestations of structural racism" (Singleton, 2022, p. 982).

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Obj. 1: Engage diverse stakeholders in the development of system dynamics models focused on implementation of community-wide food system interventions. *100% complete through establishment of a Core Modeling Team, creation of community researcher role, and implementation of Food Systems Change Fellowship.*

Obj. 2: Standardize existing data and collect and link new data to allow for empirical analysis

over short and medium time horizons and simulation analysis of long-term end points. *100% complete as evidenced by 13 scientific publications that provide empirical support for the systems modeling conducted by our team. We are finalizing a manuscript that will provide a systems science approach to data visualization for community problem-structuring with a food systems lens.*

Obj. 3: Use qualitative and visual mapping methods to build causal loop models mapping interdependencies, uncertainties, and consequences of multiple food system transformations including the role of other systemic forces (i.e., transportation). *100% complete with findings shared in 2 peer-reviewed publications and the Menu of Actions.*

Obj. 4: Empirically test, validate, and refine system dynamics models that will generate simulated output for evaluating the collective impact of community-wide food systems strategies on nutrition and food security and neighborhood economic development. *100% complete. Findings synthesized into our Menu of Actions for Community Driven Food Systems Change, released in November 2020, designed for use by lay audiences.*

Obj. 5: Develop and pilot test methods to translate the computational tool for use by community stakeholders in three neighborhoods to guide future food systems initiatives. *100% complete. We implemented the novel Food Systems Change Fellowship in 2021 with five community leaders. Our approach is synthesized into a Food Systems Change Vision Board Toolkit, released in November 2021, designed for community audiences.*

Obj. 6: Disseminate findings with diverse audiences locally, regionally, and nationally. *100% complete through 13 journal publications, 1 book chapter, 2 toolkits, 4 data briefs, and 10 public forums attracting diverse audiences.*

A summary of publications can be found in [the foodNEST 2.0 final report](#).

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

In Years 1-2, there were no changes in major goals or specific aims or milestones for this reporting period. In Years 3-4, the COVID-19 pandemic had an impact on the work of the team. We expanded our research through a FFAR supplemental grant to study the impact of COVID-19 on food security in six cities in Cuyahoga County, Ohio. Given limitations related to physically distancing, we changed all of our interactions with community stakeholders and each other to a virtual format. Additionally, we revised the approach for study dissemination to engage five community leaders as Food Systems Change Fellows in place of community-engaged sharing with leaders in the targeted study neighborhoods. There were a few delays in advancing the study timeline because of the pandemic, but all

in all the study team was able to maintain pace and productivity throughout Year 3-4.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

The end goal of this multi-year study was to develop credible and relevant tools to offer a systems lens within an equity framework to guide future local food systems change. Established via the "Tipping Points" mechanism of the Foundation for Food and Agriculture Research (FFAR) with matching support from more than a dozen partners, our study continued to serve as an organizing tool for rethinking food systems change.

YEAR 1

Aim 1: Form a core modeling team of community stakeholders who will provide guidance on model development, refinement, and dissemination.

The Core Modeling Team includes 20 community stakeholders representing different parts of the local food system. The Core Modeling Team members are paid to engage on the project either through organizational matching commitments or study provided stipends. During Year 1, members of the Core Modeling Team participated in a half-day orientation, a 2-day retreat, and in four bi-monthly meetings as of February 2019. All Core Modeling Team members met with the program manager to discuss goals and expectations and inform the development of principles of partnership to guide team interaction. A total of 17 principles were agreed upon during year 1, these were informed by prior work conducted by Community Campus Partnerships for Health. An example principle is: The relationship between partners in the Partnership is characterized by mutual trust, respect, genuineness, and commitment. Each principle is reflected upon at Core Modeling Team meetings to promote accountability and trust among the team. Additionally, the Core Modeling Team started planning the first public convening to share results of the modeling project with the public at large. The event title and goals were established, marketing plan was developed, and partners and logistics were secured. The event, Convening Ideas and Actions: Creating Food Opportunities for the Future, took place on May 22, 2019 from 8:30am – 4:00pm in Cleveland, Ohio. Members of the Core Modeling Team include:

1. Laura Adiletta, Co-Founder, Farm Fare
2. Marilyn Burns, Resident Leader and Advisor Partner
3. Devin Cotten, Program Coordinator, Burten, Bell, Carr Development, Inc.
4. Sayan Chatterjee, Professor, Case Western Reserve University
5. Kirstin Craciun, Director of Community Outreach for The Center for Health Affairs
6. Jarrod Dalton, Assistant Professor, Cleveland Clinic
7. Piper Fernwey, Community Programs & Sustainability Support Manager, Bon Appétit



Management Company

8. Margaret Fitzpatrick, Agriculture and Natural Resources Educator, The Ohio State University Extension
9. Janet Kramer, Clinical Dietician, University Hospitals, Cleveland Medical Center
10. Heather McMahon, Director of Community Health, Old Brooklyn CDC
11. Erika Meschkat, Sustainability Manager, Mayor's Office of Sustainability
12. Kristen Mikelbank, Planning Analyst Greater Cleveland Food Bank
13. Jodi Mitchell, Community Relations Manager, Perfectly Imperfect Produce
14. Amanda Osborne, Community Development Educator, The Ohio State University Extension
15. Richey Piparinen, Director, The Center for Population Dynamics
16. Sabrina Roberts, Administrator of Health Policy and Programs Health and Human Services
17. Jennifer Scofield, Chief Executive Officer, Hunger Network of Greater Cleveland
18. Morgan Taggart, Co-Chair Elect, Network of Community Advisors
19. La Queta Worley-Bell, Neighborhood Ambassador and Outreach Coordinator
20. Ana Claudia Zubieta, Program Director, Supplemental Nutrition Assistance Program

Aim 2: Use a collaborative approach to develop casual models capturing nuances of supply and demand dynamics triggered by food system tipping points with attention to mechanisms of interaction and adaptation among people (agents) and places (structures) in low resource neighborhoods.

Support from the Foundation for Food and Agriculture Research was used to develop sequentially related casual loop diagrams using a participatory system dynamic modeling approach. The causal loop diagrams map the complexity of supply and demand dynamics triggered by food system tipping points with attention to mechanisms of interaction and adaptation among people (agents) and places (structures) in low resource neighborhoods. As of Year 1, the team developed a causal loop diagram of the broader forces influencing food system dynamics as well as three models mapping the system dynamics of targeted tipping points in the food system:

1. Use of Emergency Food Assistance,
2. Development of Healthy Food Access in Neighborhoods, and
3. Use of Healthy Food Incentive Programs.

The goal in all of these models is to unpack the dynamics influencing the three main outcome variables of the study: food security, economic opportunity, and nutrition equity. Additionally, members of the Core Modeling Team, working in partnership with the investigators, have started the process of describing and operationalizing the "preferred state" of the system in terms of the primary outcomes. In other words, the team is deliberating what needles should move to satisfy local decision makers about the effectiveness of a strategy in terms of improving the three main outcomes. These endpoints are critical as we move into the next phase of the study involving simulation modeling of "what if" scenarios that allow for testing of different interventions based on the causal loop diagrams developed in year 1.

Aim 3: Build an integrated platform to empirically evaluate the collective impact of multiple food system tipping points on the neighborhood food retail environment; residential food norms dietary behaviors, and health risk, and neighborhood economic development. This



includes longitudinal individual level data related to food habits and diet as well as food retail data combined with publicly available data (e.g., census) and data from our partners.

The focus in Year 1 was on collecting the longitudinal data. To date, 303 surveys were completed (172 from Cleveland and 131 from our comparison community in Columbus). Additionally, 69 food stores were audited to identify quality, variety, price of healthy foods at retail outlets in three neighborhoods in Cleveland that are the targets for the systems modeling project. We met with our partners from the Greater Cleveland Food Bank, Hunger Network of Greater Cleveland, Ohio State University Extension, and Ohio Nutrition Incentive Network met with our team members to discuss their data systems that will be used as a part of the data platform. These partners will contribute real world implementation data that will enhance the validity of our modeling because their data reflect true implementation of our targeted tipping points

Aim 4: Simulate the impact of tipping points over longer time horizons and in different neighborhoods to identify intended and unintended consequences, inflection points, interactions, and outcomes with the goal of applying this information to inform future food system action planning.

Aim 4 not yet addressed in Year 1.

YEAR 2

Aim 1: Form the Core Modeling Team with experts from local community groups working in food systems and research institutions who will offer guidance on model development, refinement, and dissemination.

After two years of work, our local team was increasingly interested in using lessons from the systems modeling process to inform implementation of select food system transformations that are emerging from foodNEST 2.0's participatory system dynamic modeling, which is propelling the team to think critically about translating the modeling insights into funding, operational and logistical, and evaluation strategies that align with a systems framework and equity lens to support activation of this plan. A key lesson from 2019 is related to our conceptualization of "tipping points" in the food system that led to a preferred state of economic opportunity, food security, and nutrition equity in historically redlined neighborhoods.

Aim 2: Use a collaborative approach to develop and refine causal models capturing nuances of supply and demand dynamics triggered by food system tipping points with attention to mechanisms of interaction and adaptation among people (i.e., "agents," like producers and consumers) and places (i.e., "structures," like new food business development and emergency food assistance) in racialized neighborhoods.

Based on engagement of >250 stakeholders in the participatory system dynamic modeling efforts, tipping points for equitable food systems change are being redefined. The revised set of levers (still under development as of Year 2) include household financial strength, gentrification, coordination of resources, community ownership of neighborhood change, political engagement, incarceration and parole, farmer vitality, and culture, values, and norms. These tap into different levels of systems change notably moving beyond parameter changes (i.e., adding 1+ food pantry or supermarket in the neighborhood) to target changes in feedback (i.e., structures of racism that limit the rate change), design (i.e., change rules of

the game to incentivize equity as an outcome), and intent (i.e., goal of the food system is to promote economic opportunity, food security, and nutrition equity) of the system. These have implications for the types of food systems transformation strategies that will emerge from our models as well as the structures needed to support implementation. These insights emerged through qualitative causal loop diagramming that mapped out interdependencies of food systems dynamics and application of these ideas through diverse community forums to begin to scope out solutions that are systems-oriented and equity-focused.

Aim 3: Build an integrated data platform to empirically evaluate the collective impact of food system tipping points on the neighborhood food retail environment; residential food norms, dietary behaviors, and health risk; and neighborhood economic development.

Another main outcome from Year 2 was the identification of key stakeholder groups that are perceived to have power to pull the levers for food systems transformation (i.e., residents, retailers, and regulators). During the latter part of Year 2, the study team developed three quantitatively informed simulators (i.e., stock-and-flow models) that served as a tool for testing the impact of hypothesized food systems solutions.

Aim 4: Simulate the impact of tipping points over longer time horizons and in different neighborhoods to identify intended and unintended consequences, inflection points, interactions, and outcomes with the goal of applying this information to inform future food system action planning.

Feedback on the simulators described in Aim 3 is being used to refine the models to align simulator parameters to test specific food systems solutions (i.e., increase in SNAP benefits + changes in living wage policy). These will be revealed publicly in Year 3.

Finally, the modeling efforts have reinforced the need for our team to devote additional research into the ideal organizational structure for food systems change. It became increasingly clear that this structure for shared planning, programming, policy, community engagement, and data does not exist in the local food system. We decided to hire a consultant with expertise in organizational change to facilitate deeper discussions about the ideal organizational structure to support coordinated and integrated food systems change.

YEAR 3

Aim 1: Form the Core Modeling Team with experts from local community groups working in food systems and research institutions who will offer guidance on model development, refinement, and dissemination.

In Year 3, with support from two external consultants, the team went through a process of refining the goals of our collaboration as we began moving away from focusing explicitly on model building into a phase of model translation, application, and refinement. A new meeting structure was organized with three meetings per month open to all members of the collaborative. In the past years, we held a core modeling team meeting bi-monthly and internal study team meetings three times per month. The new model sought to continue to engage all stakeholders in the work of the study through two "Core Meetings" per month focused on modeling and data as well as our supplemental grant addressing food security and COVID-19. Once per month, we held a "Collaborative Meeting" for all interested in current study goals as well as plans for expanding the work in the future. In general, about 15-30

people attend these weekly meetings. Additionally, we created a new role on the study for community members engaged with the Core Modeling Team for at least 12 months. This new role, Community Researcher, was intended to replace the formal role of Core Modeling Team member.

Aim 2: Use a collaborative approach to develop and refine causal models capturing nuances of supply and demand dynamics triggered by food system tipping points with attention to mechanisms of interaction and adaptation among people (i.e., "agents," like producers and consumers) and places (i.e., "structures," like new food business development and emergency food assistance) in racialized neighborhoods.

Results of our modeling work revealed nutrition equity as a high-level goal of food systems in historically redlined neighborhoods. Nutrition equity is defined as households healthy in body, mind, and spirit with freedom, agency, and dignity over their food traditions. Three domains of feedback were identified related to meeting basic food needs with dignity, market-based supply and demand for fresh and healthy foods, and community empowerment and food sovereignty. Within these domains, 10 feedback mechanisms were identified. The team also identified moderating factors that are exogenous to neighborhood-based food systems including neighborhood crisis, neighborhood investment for racial equity, other household costs, funding for government benefits, and voter participation.

Aim 3: Build an integrated data platform to empirically evaluate the collective impact of food system tipping points on the neighborhood food retail environment; residential food norms, dietary behaviors, and health risk; and neighborhood economic development.

Building on findings from our study (evidenced through the five papers accepted for publication in Year 3), we worked to finalize the "data pantry" prototype that will provide metrics for simulating food systems changes in historically redlined neighborhoods. Using insights from the Food Systems Change Fellowship program, we planned to empirically evaluate the utility of this data platform and the resulting simulation models based on insights from our core model of nutrition equity established in Aim 2.

Aim 4: Simulate the impact of tipping points over longer time horizons and in different neighborhoods to identify intended and unintended consequences, inflection points, interactions, and outcomes with the goal of applying this information to inform future food system action planning.

It became increasingly clear that simulation modeling can be most effectively applied to study how best to implement and scale up promising strategies that improve neighborhood food security and access to fresh and healthy foods. As a part of the Fellowship program, we will work with Fellows to develop strategy-specific simulators to guide action planning for their transformation proposition. These simulators will build on the models developed in Aim 2 and the data platform created in Aim 3.

YEAR 4

Aim 1: Form the Core Modeling Team with experts from local community groups working in food systems and research institutions who will offer guidance on model development, refinement, and dissemination.

Collaborative meetings continued three times per month to support the dissemination and

translation of key foodNEST 2.0 findings, refine and conclude all modeling work, and continue informing the direction of the food security and COVID-19 Supplement. In the summer of 2021, these meetings expanded to include any community partner interested in learning about and/or sharing foodNEST 2.0 findings not just those on the Core Modeling Team. The collaborative supported the development of an op-ed published in The Plain Dealer and the planning of three public convenings: two mayoral candidate conversations and the foodNEST 2.0 final dissemination event, foodNEST & friends. Four Community Researchers joined the study team from April 2021 through December 2021, each providing 10 hours of consulting per month, to ensure that dissemination efforts and findings were relevant and accessible to target stakeholders in the community.

Aim 2: Use a collaborative approach to develop and refine causal models capturing nuances of supply and demand dynamics triggered by food system tipping points with attention to mechanisms of interaction and adaptation among people (i.e., "agents," like producers and consumers) and places (i.e., "structures," like new food business development and emergency food assistance) in racialized neighborhoods.

The causal loop diagram resulting from the work of the Core Modeling Team was finalized in Year 4. This diagram, as well as the other insights from the group, resulted in the 2 publications. One of which - "Food system dynamics structuring nutrition equity in racialized urban neighborhoods" published in the American Journal of Clinical Nutrition – received particularly strong support in the form of a concurrently published editorial piece advocating on behalf of our findings. Furthermore, the insights from the causal models were translated into the Menu of Actions, which translates these insights into language accessible by key stakeholder audiences.

Aim 3: Build an integrated data platform to empirically evaluate the collective impact of food system tipping points on the neighborhood food retail environment; residential food norms, dietary behaviors, and health risk; and neighborhood economic development.

Interdisciplinary research in complex fields such as public health and food systems must assess multiple domains such as neighborhood economic development, dietary behaviors and norms, and the neighborhood food retail environment to support transparent and replicable modeling of food systems. However, empirically grounded meaningful metrics that comprehensively represent such complex phenomena rarely exist in data dashboards that are accessible to both system dynamics modelers and public health data managers.

We are finishing up a manuscript stating how data dashboards should be designed to reduce research waste and improve food system insights. As no synergistic standardization method exists to capture data that is meaningful and conducive to system dynamics work, our perspective piece will initiate a conversation on how both traditional data management and system dynamics approaches can work synergistically. This manuscript will include recommendations on how to organize data to support our food system dynamics structuring framework (Freedman et al., 2021). Affirming that there is no one way to accomplish this, we elaborate on the fundamental concept of a 'reference mode', as applied within system dynamics modeling, to guide the design of a generalizable informatics platform (i.e., dashboard) that enables variable harmonization, variable aggregation-disaggregation, and variable rates of change over time (i.e., delays, fractional rates, incidence rates). We intend to submit this work for peer review and publication in Frontiers in Public Health in 2022.

The core elements contained in the foodNEST 2.0 “data pantry” include NDSR 24-hour recall data, foodNEST survey questions, and external data sources. We have created a metadata file to describe these data, and we have pdf data summaries for each of the five years of survey data. These pdf files link variable names to summary statistics and the exact wording of questions and responses. The foodNEST survey data for each of the five years are saved in Excel files.

Aim 4: Simulate the impact of tipping points over longer time horizons and in different neighborhoods to identify intended and unintended consequences, inflection points, interactions, and outcomes with the goal of applying this information to inform future food system action planning.

Given limitations related to physically distancing, we changed all of our interactions with community stakeholders and each other to a virtual format. As such, we revised the approach for simulation testing from a broad-reaching community-engaged sharing with leaders in the targeted study neighborhoods to a deeper form of engagement with five community leaders via the Food Systems Change Fellowship. The approach of the Food Systems Change Fellowship is synthesized into a Food Systems Change Vision Board Toolkit, released in 2021, and is designed for community audiences.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Multiple steps were taken to ensure robust and unbiased research approaches. First, we engaged an interdisciplinary team of investigators with backgrounds in population health, epidemiology, community psychology, food policy, nutrition, social work, and community development. This ensured diverse perspectives, theories, and epistemologies underpinned the study approach from design to dissemination. Second, we engaged community partners as co-creators of the research relying on their active engagement in all phases of the participatory system dynamic modeling processes. These individuals represented diverse perspectives related to the food system and included nutrition and agriculture educators, food service and emergency food assistance providers, grassroots food justice advocates, healthcare stakeholders, nonprofit and local government leaders, community leaders, and people with lived experiences navigating food systems in historically redlined neighborhoods. Third, we employed mixed methods to try to understand the complex system dynamics structuring food systems in racialized urban neighborhoods. This allowed for triangulation of findings from both qualitative and quantitative methods and from data collected with different groups. Fourth, all data collection and analysis processes were standardized to promote rigor and reproducibility. Whenever possible, we used or adapted existing tools for data collection to align with best practices in the field. Standard operating procedures were created, and study staff were trained and monitored to ensure fidelity. Quality assurance processes were implemented to assess data quality as well as replicability of data analysis. Fifth, results of our research were submitted to journals for scientific peer review allowing external audiences to

assess the quality and rigor of our research and offer feedback for improvement. The study team published 13 papers in peer reviewed journals during the study timeframe.

2.8. List key stakeholders that could be served by results of this project. Key stakeholders that can be served by the results of this project include policy makers, food policy coalitions, (urban) farmers, food hubs, food retailers, hunger relief and food access groups, community leaders, government agencies, social service providers, health care and public health partners, private sector business, consumers, and researchers.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

The foodNEST 2.0 team utilized the following strategies and diverse methods for translation in Year 4 to disseminate findings to communities of interest. It is worth noting that all of our collaborative study meetings and the Fellowship workshops include dissemination of study findings for feedback, deliberation, reflection, and integration into action planning.

1. Study website with comprehensive information about the study overall including our COVID-19 supplemental project. This is embedded in the Mary Ann Swetland Center for Environmental Health website at CWRU.
2. "Vision Board Toolkit" - The toolkit summarizes our process for implementing the Food Systems Change Fellowship as well as a step-by-step process for moving food systems change visions into action.
3. Op-ed: "A new Cleveland mayor, a new opportunity for food justice: Michelle B. Jackson and Gwendolyn Garth" - Op-ed published in the Cleveland Plain Dealer and Cleveland.com. Our study team convened a group of community leaders to conceptualize, write, and publish the op-ed calling the new Cleveland mayor into food justice.
4. Public Convenings:
 - a. "Showcase of Opportunities" - hybrid public convening, July 28, 2021. Showcased visions for food systems change from five Food Systems Change

- Fellows. A total of 141 attendees joined virtually or in person.
- b. "Setting the Table for Food Policy: A conversation with mayoral candidate Kevin Kelley" - virtual public convening co-hosted by foodNEST 2.0 with the Cleveland-Cuyahoga County Food Policy Coalition and Ohio Policy Matters, October 15, 2021. A total of 39 attendees joined the virtual convening.
 - c. "Setting the Table for Food Policy: A conversation with mayoral candidate Justin Bibb" - virtual public convening co-hosted by foodNEST 2.0 with the Cleveland-Cuyahoga County Food Policy Coalition and Ohio Policy Matters, October 19, 2021. A total of 41 attendees joined the virtual convening.
 - d. "fooNEST & friends" - virtual public convening, December 1, 2021. Featured breakout sessions for action planning with foodNEST 2.0 authors paired with Food Systems Fellowship alumnae. A total of 79 attendees joined the virtual convening.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Year 1 - While the study funding began April 1, 2018, it took us about three months to recruit and hire core team members including our full-time program manager, post-doctoral scholars, and pre-doctoral scholar. This resulted in some delays in using funds from the FFAR Tipping Points grant.

Year 2 - One main issue related to the study in Year 2 was balancing the tension between (a) thinking through the complexity of the food system and using this information to inform our system dynamic models, which takes time, and (b) acting to implement food system changes now because the needs are high and many decades in the making. Some partners are concerned that we have been talking a lot about the models and they are ready to act. This is a valid concern; however, congruent with the multi-year plan for the entire foodNEST 2.0 study, our models are not yet validated to the point where we feel they are ready to act upon.

Year 3 - The biggest challenge in Year 3 was the COVID-19 pandemic, which affected every part of society from our personal lives, to our working approach, to our research process. In May 2020, we held a public convening with the Cleveland-Cuyahoga County Food Policy Coalition to get feedback from policy makers and community leaders on the impact of COVID-19 on food security. We also conducted assessments with our team and partners to find ways to engage virtually. This transition was successful; however, it did require provision of additional technology supports (i.e., computer, hotspot) to some of community partners to facilitate full engagement via or new virtual format. Lastly, COVID-

19 impacted how we would be are sharing and applying study findings in the community (a key goal in Year 3 of our grant). It was not possible to meet face-to-face with communities through prior methods such as attendance at community meetings. Therefore, we shifted our approach for community translation in the format of a fellowship program designed to support community leaders interested in advancing food systems change in their neighborhoods.

Year 4 – In Year 4, the biggest challenge continued to be the COVID-19 pandemic, which largely affected the implementation of the Foods Systems Change Fellowship and dissemination activities. The 2021 Food Systems Change Fellowship, implemented from February through June 2021, was designed and implemented in a completely virtual environment aside from Showcase practice sessions and the Showcase itself. This presented technical challenges (e.g. Internet, equipment) and created barriers for relationship building. Our team was able to overcome relationship building barriers by hosting office hours, being available by phone, and providing 1:1 support to Fellows. The pandemic's influence persisted in our dissemination. The team was unable to physically attend conferences (all presentations were remote), and Case Western Reserve University prohibited large, in-person gatherings. This necessitated that our final dissemination be virtual. We are pleased that over 70 attendees joined us on Zoom for this final "foodNEST & friends" event.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Year 4 – 04/01/2021-12/31/2021

We trained two Post-Doctoral and three Pre-Doctoral Scholars. Notably, 60% of these trainees are Black, Indigenous and People of Color reflecting our commitment to train the next generation of environmental health scholars representing diverse backgrounds.

Post-Doctoral Scholars

1. Eun Kyung Lee, PhD, MPH joined the team in January 2020 after completing her PhD in environmental health sciences at University of Albany. Eun Kyung provides leadership to data documentation procedures that will support replication of the modeling efforts. Eun Kyung completed her Post-doc with the Swetland Cener in January 2022 and will continue on as a postdoctoral research associate with the Boston University School of Public Health in the Dept. of Health Law, Policy, and Management.



2. Owusua Yamoah, PhD joined the team in May 2020 after completing her PhD in integrated social and spatial sciences from University of Toledo. With a background in community development, qualitative analysis, and spatial sciences, Owusua provides leadership to the Food Security and COVID-19 supplement awarded to our team in summer 2020. She received a pilot grant from NIH, through the Center for Reducing Health Disparities, for her proposal to assess spatial and temporal variability in emergency food assistance use. Owusua completed her post-doctoral position in April 2022 and was selected as a full-time Research Scientist within the Swetland Center in April 2022.

Pre-Doctoral Scholars

3. Gwendolyn Donley, PhD, MPH completed her PhD requirements in Epidemiology and Biostatistics at CWRU as of May 2022. She was accepted as a FFAR Fellow at NC State University in 2019. Gwen's dissertation research builds on insights from foodNEST 2.0 with a focus on examining the impact of housing instability on perceptions of the food environment and COVID-19 vaccine acceptance. Gwen defended her dissertation in March 2022. She was selected for the Presidential Management Fellowship and accepted a position as a National Poverty Fellows post-doctoral fellow at the Office of Community Service (OCS) with the federal Administration for Children and Families.

4. India Gill, MPH is a doctoral candidate in Epidemiology and Biostatistics at CWRU. She joined the team in January 2020 to support the development of community-engagement strategies to translate the food system simulators for use by stakeholders at the neighborhood level. India provides core support to the Food Systems Change Fellowship launched in Year 3 of our grant. India received a grant from the Flore Stone Mather Center for Women to fund a portion of her dissertation focused on investigating neighborhood level racial inequities in child asthma exacerbations. She plans to complete all degree requirements by August 2022.

5. Abigail Roche, MS, RD is a doctoral candidate in Epidemiology and Biostatistics at CWRU. She has a background in nutrition and dietetics and has experience working in food as medicine programming focused on people living with HIV. Abigail was accepted as a FFAR Fellow at NC State University in 2020. She provided leadership to the Modeling Core of the project and is assisting with qualitative data analysis related to our COVID-19 and food security supplemental award.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Year 4 – 04/01/2021-12/31/2021

Post-doc: 2

Pre-doc: 3

Graduate: 4

Undergraduate: 2

During Year 4, support from FFAR allowed us to create 11 positions for students and trainees.

Of which, 2 are Post-Doctoral Scholars: Owusua Yamoah, PhD, MA and Eun Kyung Lee, PhD, MPH. Three are Pre-Doctoral scholars: Gwendolyn Donley, MPH; Abigail Roche, MS; and India Gill, MPH – all of whom are doctoral students in Epidemiology and Biostatistics at CWRU. Additionally, we supported 4 graduate students. Audrey Lim, BS, MS, who completed her Master of Science in Spring 2021, supported data collection for the food security and COVID-19 supplement. Maurice Lucious, working toward a Master of Public Health at Walden University, supported the planning of the foodNEST & friends dissemination day. Rebecca Schulte, working towards a Master of Public Health-Epidemiology at Colorado State University, supported the implementation of the Food Systems Change Fellowship. Finally, Thiago Gonzalez holds a BS in Engineering and is finalizing his Masters in Environmental Science. Thiago supported tasks related to the conclusion of foodNEST 2.0 and is working with the team in the development of Nourishing Neighborhoods, Empowering Communities. Finally, we are pleased to have supported two undergraduate students in this reporting period. Katie Merritt is an undergraduate from CWRU working with Co-I Jill Clark on an independent study, and Rachel Wyetzner is an undergraduate from CWRU who supports dissemination and communications efforts for foodNEST 2.0.

3. Information Products - DONE

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scholarly Publication

Toolkit

Data Briefs

Book Chapter

Public Convenings

Conference Presentations

Invited Presentations

Published op-ed

- 3.2. Please provide a list of citations for the information products produced during the project.

Scholarly Publications

1. Pike, S. N., Trapl, E. S., Clark, J. K., Rouse, C. D., Bell, B. A., Sehgal, A. R., To, T., Borawski, E., & Freedman, D. A. (2017). Examining the Food Retail Choice Context in Urban Food Deserts, Ohio, 2015. *Preventing Chronic Disease*, 14, E90.
2. Freedman, D. A., Bell, B. A., Clark, J. K., Sharpe, P. A., Trapl, E. S., Borawski, E. A., Pike, S. N., Rouse, C., & Sehgal, A. R. (2019). Socioecological Path Analytic Model of Diet Quality among Residents in Two Urban Food Deserts. *Journal of the Academy of Nutrition and Dietetics*, 119(7), 1150–1159.
3. Freedman, D.A., Bell, B., Clark, J., Ngendahimana, D., Borawski, E., Trapl, E., Pike, S., & Sehgal, A.R. (2020). Small improvements in an urban food environment resulted in no changes in diet among residents. *Journal of Community Health*.
4. Gupta, Nisha (2020). Food Security Moderates Relationship between Perceived Food Environment and Diet Quality among Adults in Communities with Low Access to Healthy Food Retail. *Journal of Public Health Nutrition*.
5. Banks, AR, Bell, BA, Ngendahimana, D, Embaye, M, Freedman, DA, Chisolm, DJ. (2021). Identification of factors related to food insecurity and the implications for social determinants of health screenings. *BMC Public Health*.
6. Benninger, E., Donley, G., Schmidt-Sane, M., Clark, J.K., Lounsbury, D.W., Rose, D. and Freedman, D. (2021). Fixes that Fail: A system archetype for examining racialized structures within the food system. *Am J Community Psychol*.
7. Chisolm, DJ, Bell, BA, Yamoah, O, & Freedman, DA. (2021). Racialized experiences differentiate food security among African American adults. *Nursing Research*, 70(5), S13-S20.
8. *Ciesielski, T.H., Ngendahimana, D.K., Roche, A., Williams, S.M., Freedman, D.A. (2021). Elevated dietary inflammation among SNAP recipients provides targets for precision public health intervention. *American Journal of Preventive Medicine*.
9. *Glickman A., Clark J.K., & Freedman D. (2021). A relational approach to the food environment finds that the food environment matters for those who use it. *Health & Place*.
10. Jablonski, B.B.R., Casnovsky, J., Clark, J.K., Cleary, R., Feingold, B., Freedman, D., Gray, S., Romeiko, X., Olabasi, L.S., Torres, M., van den Berg, A.E., Walsh, C., & Wentworth, C. (2020). Emergency food provision for children and families during the COVID-19 pandemic: Examples from five U.S. cities. *Applied Economic Perspectives and Policy*.
11. Sanchez-Flack, J., Joshi, K., Lee, E.E., & Freedman, D.A. (2021). Indicators of Readiness and Capacity for Implementation of Healthy Food Retail Interventions. *Journal of Agriculture, Food Systems, and Community Development*.
12. Freedman, D.A., Clark, J.K., Lounsbury, D.W., Boswell, L., Burns, M., Jackson, M.B., Mikelbank, K., Donley, G., Worley-Bell, L., Mitchell, J., Ciesielski, T.H., Embaye, M., Lee, E.K., Roche, A., Gill, I., & Yamoah, O. (2022). Food system dynamics structuring nutrition equity in racialized urban neighborhoods. *American Journal of Clinical Nutrition*.
13. Lee, E. K., Donley, G., Ciesielski, T. H., Gill, I., Yamoah, O., Roche, A., Martinez, R., Freedman, D. A. (2022). Health Outcomes in redlined versus non-redlined neighborhoods: A systematic review and meta-analysis. *Social Science & Medicine*. 294, 114696.

Toolkits

1. Modeling the Future of Food in Your Neighborhood Collaborative. (November 2020). Menu of Actions for Community Driven Food Systems Change. Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, OH.
2. Boswell, L., Burns, M., Clark, J., Donley, G., Etheridge, K., Freedman, D., Fryerson, S., Gill, I., Harris, A., Hovmand, P., Jackson, M. B., Kahl, B., Lounsbury, D., Sommer, R. (September 2021). The Food Systems Change Vision Board Toolkit. Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, OH.

Data Briefs

1. Donley G, Arima G, Clark J, Trapl E, Freedman D. Healthy Food Availability in Four Ohio Neighborhoods: Findings from the Nutrition Environment Measures Surveys, 2015-2019. (September 2020). Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, Ohio.
2. Modeling the Future of Food in Your Neighborhood Collaborative. (November 2020). Executive Summary: Menu of Actions for Community Driven Food Systems Change. Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, OH.
3. King, J., Rose, D., Burt, N., Taggart, M., Mills, F., & Freedman, D. (September 2019). Convening Ideas & Actions: Creating Food Opportunities for the Future - 2019 Convening Findings. Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, Ohio.
4. Yamoah, O., Roche, A., Athreya, B., Kahl, B., Lim, A., Mikelbank, K., Freedman, D. (2021). Food Security and COVID-19 in Cuyahoga County: Common Shock, Varied Impact by City. Mary Ann Swetland Center for Environmental Health, Case Western Reserve University, Cleveland, Ohio.

Book Chapter

1. Glickman, A.R., Clark, J.K., Freedman, D.A. (2022). Participatory modeling of the food system: The case of community-based systems dynamics. In: C. Peters (Ed.). Food Systems Modelling, 257-283. Academic Press, Cambridge, MA.

Public Convenings

1. "Convening Ideas and Actions: Creating Food Opportunities for the Future." - Public convening co-organized with Cleveland-Cuyahoga County Food Policy Coalition, Food Access Raises Everyone, Health Improvement Project-Cuyahoga, and Cleveland Office of Minority Health, May 22, 2019.
2. "Update on COVID-19 and Food Policy in Greater Convening" - virtual public convening co-organized by the Study with the Cleveland-Cuyahoga County Food Policy Coalition, May 29, 2020. Featured local leaders including a member of Cleveland City Council, local chef, and president of the local food bank. A total of 58 attendees joined the webinar.
3. "Count Me In: Food Policy, Census, and Voting During COVID-19" - virtual public convening co-organized by the Study with the Cleveland-Cuyahoga County Food Policy Coalition, September 11, 2020. Panelists included three members of our Core Modeling Team and two members of the food policy coalition. A total of 65 attendees joined the webinar.



4. "Menu of Actions for Community Driven Food Systems Change" – virtual panel presentation, November 17, 2021. Panelist included members of the study team and Core Modeling Team including a community advocate, pre-doctoral scholar, public health leader, community gardener, and co-investigator. A total of 49 attendees joined the webinar.
5. "Showcase of Opportunities" - hybrid public convening, July 28, 2021. Showcased visions for food systems change from five Food Systems Change Fellows. A total of 141 attendees joined virtually or in person.
6. "Setting the Table for Food Policy: A conversation with mayoral candidate Kevin Kelley" virtual public convening co-organized by the Study with the Cleveland-Cuyahoga County Food Policy Coalition and Ohio Policy Matters, October 15, 2021. A total of 39 attendees joined the virtual convening.
7. "Setting the Table for Food Policy: A conversation with mayoral candidate Justin Bibb" virtual public convening co-organized by the Study with the Cleveland-Cuyahoga County Food Policy Coalition and Ohio Policy Matters, October 19, 2021. A total of 41 attendees joined the virtual convening.
8. "fooNEST & friends" - hybrid public convening, December 1, 2021. Featured breakout sessions for action planning with foodNEST 2.0 authors paired with Food Systems Fellowship alumnae. A total of 79 attendees joined the virtual convening.

Conference Presentations

Year 1

1. Freedman, D. (2018). Integrating systems science into efforts to promote nutrition equity. Oral presentation conducted at the Prevention Research Center for Healthy Neighborhoods Seminar Series. Cleveland, Ohio.
2. Donley, G., Benninger, E., Rose, D., Lounsbury, D., Clark, J., Trapl, E., Bush, W., & Freedman, D. on behalf of the Modeling the Future of Food in Your Neighborhood Team (2019). Using participatory systems dynamic modeling to assess nutrition equity in low resource neighborhoods. Poster presented at: The Society for Applied Anthropology 79th Annual Meeting; 2019 March 19-23; Portland, OR.

Year 2

1. Benninger, E., Rose, D., Lounsbury, D., Clark, J., Trapl, E., Bush, W., & Freedman, D. (2019, May). Moving to system approaches: Building system dynamic models of local food system dynamics. In Freedman, D. (Chair), Advancing implementation science to promote nutrition equity. 20x20 presentation conducted at the Society for Prevention Research 27th annual meeting, San Francisco, CA.
2. Freedman, D.A., Lounsbury, D. Bess, K., Zimmerman, L., & Foster-Fishman, P. (2019, June). Developing training to advance community systems science for

health equity. Roundtable discussion at the Society for Community Research and Action Biennial Conference, Chicago, IL.

3. Benninger, E., Freedman, D., Lounsbury, D., Clark, J., Rose, D., Bush, W., & Trapl, E. on behalf of the Modeling the Future of Food in Your Neighborhood Team (2019, June). Understanding Neighborhood Access to Healthy Food. Poster session presented at the Society for Community Research and Action Biennial Conference, Chicago, IL.

Year 3

1. Glickman A. (2020, June). "Community-Based Systems Modeling, Public Participation & Wicked Problems: Towards A Comprehensive Framework for Understanding Public Vision" Public Administration Theory Network Annual Meeting, Virtual.
2. Benninger, E., Donley, G., Schmidt-Sane, M., Rose, D., Clark, J., Lounsbury, D., & Freedman, D. (2020, October). Systems thinking and the "fixes that fail" systems archetype as a model for conceptualizing food system inequities. Oral presentation submitted for the American Public Health Association Conference. San Francisco, Virtual.
3. Embaye, M., Freedman, D., Ngendahimana, D., Bell, B., Zaim, H., Banks, A., & Chisolm, D. (2020, October). Identifying targets for health promotion efforts designed to reduce food insecurity. Oral presentation submitted for the American Public Health Association Conference. San Francisco, Virtual.
4. Glickman A., Clark J, & Freedman D. (2020, Nov 11-13). "Residential Proximity to Low-Quality Food Retailers and Diet Behavior: Exploring the Micro Food Environment within Low-Income Neighborhoods" APPAM 2020 Annual Meeting, Virtual.
5. Donley, G., Benninger, E., Schmidt-Sane, M., Rose, D., Clark, J., Lounsbury, D., Bush, W., Freedman, D. (2021, March). Using Community System Dynamics Modeling to Identify Points of Leverage for Improving Urban Food Systems. 81st Annual Meeting for the Society for Applied Anthropology.

Year 4

1. Freedman, D.A. (2021, March). Restructuring our mental models for nutrition research. Case Comprehensive Cancer Center Mini Retreat on Behavioral Science, Cleveland, OH.
2. Ciesielski, T.H., Ngendahimana, D. Roche, A., Williams, S.M., Freedman, D.A. Research Gaps and Opportunities Workshop Nutrient Deficits Among Urban SNAP Recipients: Implications for Inflammation, Chronic Disease, Immune Function, and COVID-19. Poster presented at the Precision Nutrition, Virtual.
3. Donley G, Bush W, Curtis A, Freedman D. (2021, October) Residential instability is not associated with changes in perceived food environments despite objective changes in grocery store availability among primary food shoppers in historically redlined neighborhoods. Annual Meeting of the

Interdisciplinary Association for Population Health Sciences. Virtual/Baltimore, Maryland.

4. Yamoah, O., Roche, A., Athreya, B., Kahl, B., Lim, A., Mikelbank, K., Freedman, D. (2021, September). Municipal Responses to Food Security Resulting from COVID-19 in Cuyahoga County. Poster Presentation for the National Institutes of Health Workshop: Food Insecurity, Neighborhood Food Environment, and Nutrition Health Disparities: State of the Science, Virtual.
5. Donley, G., Benninger, E., Schmidt-Sane, M., Rose, D., Clark, J., Lounsbury, D., Bush, W., & Freedman, D. (2021, September). Poverty, Crisis, and Stress as Root Causes of Food System Inequity. Poster Presentation for the National Institutes of Health Workshop: Food Insecurity, Neighborhood Food Environment, and Nutrition Health Disparities: State of the Science, Virtual.
6. Donley, G., Benninger, E., Schmidt-Sane, M., Rose, D., Clark, J., Lounsbury, D., Bush, W., Freedman, D. (2021, March) Using Community System Dynamics Modeling to Identify Points of Leverage for Improving Urban Food Systems. 81st Annual Meeting for the Society for Applied Anthropology. Online/Norfolk, Virginia.

Invited Presentations

Year 2

1. Freedman, D.A. (2019, November). Modeling the Future of Food in Your Neighborhood Study. Invited presentation at the Smart Urban Food Systems Summit, Brooklyn, NY.

Year 3

2. Clark, JK., King, J. (2020, October). 'Fixes that Fail:' Using community-based systems modeling to diagnose injustice in the food system, Keynote presentation, Urban Food System Symposium, Kansas State University.
3. (2020, October). Systems Modeling to Advance Equitable Food Systems. Invited presentation at Albert Einstein College of Medicine. Bronx, NY/Virtual Presentation.
4. Freedman, D.A. (2021, March). Mechanisms of racism structuring food security and community nutrition. Invited presentation at the Population Health Research Institute Seminar Series, MetroHealth System, Cleveland, OH.

Year 4

5. Clark, JK. (2021, April). 'Fixes that Fail:' Using community-based systems modeling to diagnose injustice in the food system, Kirwan Institute for the Study of Race and Ethnicity Forum, Ohio State University.
6. Freedman, DA, Jackson, MB. 2021, August. Serving Up Lessons Learned from our Food Systems Modeling Journey. Invited plenary presentation at the Michigan State University Participatory Modeling Conference, East Lansing, MI. (Virtual Presentation)

7. Freedman, DA. 2021, December. Food insecurity needs and opportunities in Cuyahoga County. Healthy Northeast Ohio Collaborative, Center for Health Affairs, Cleveland, OH.
8. Freedman, DA. 2022, February. Mechanisms of Racism Structuring Food Security and Community Nutrition. Goldbag Research Seminar, Case Western Reserve University, School of Nursing, Cleveland, OH.
9. Freedman, D.A. (2022, March). Mechanism Structuring Nutrition Equity in Neighborhood-based Food Systems. Invited presentation at Tufts University Friedman School of Nutrition Science and Policy. Medford, MA (Virtual presentation).

Op-ed

2. (2021, November 21). A new Cleveland mayor, a new opportunity for food justice: Michelle B. Jackson and Gwendolyn Garth. *The Plain Dealer and Cleveland.com*. <https://www.cleveland.com/opinion/2021/11/a-new-cleveland-mayor-a-new-opportunity-for-food-justice-michelle-b-jackson-and-gwendolyn-garth.html>.
- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Peer-Reviewed Publications

1. Pike, S. N., Trapl, E. S., Clark, J. K., Rouse, C. D., Bell, B. A., Sehgal, A. R., To, T., Borawski, E., & Freedman, D. A. (2017). Examining the Food Retail Choice Context in Urban Food Deserts, Ohio, 2015. *Preventing Chronic Disease*, 14, E90.
2. Freedman, D. A., Bell, B. A., Clark, J. K., Sharpe, P. A., Trapl, E. S., Borawski, E. A., Pike, S. N., Rouse, C., & Sehgal, A. R. (2019). Socioecological Path Analytic Model of Diet Quality among Residents in Two Urban Food Deserts. *Journal of the Academy of Nutrition and Dietetics*, 119(7), 1150–1159.
3. Freedman, D.A., Bell, B., Clark, J., Ngendahimana, D., Borawski, E., Trapl, E., Pike, S., & Sehgal, A.R. (2020). Small improvements in an urban food environment resulted in no changes in diet among residents. *Journal of Community Health*.
4. Gupta, Nisha (2020). Food Security Moderates Relationship between Perceived Food Environment and Diet Quality among Adults in Communities with Low Access to Healthy Food Retail. *Journal of Public Health Nutrition*.
5. Banks, AR, Bell, BA, Ngendahimana, D, Embaye, M, Freedman, DA, Chisolm, DJ. (2021). Identification of factors related to food insecurity and the implications for social determinants of health screenings. *BMC Public Health*.
6. Benninger, E., Donley, G., Schmidt-Sane, M., Clark, J.K., Lounsbury, D.W., Rose, D. and Freedman,

- D. (2021). Fixes that Fail: A system archetype for examining racialized structures within the food system. *Am J Community Psychol*.
7. Chisolm, DJ, Bell, BA, Yamoah, O, & Freedman, DA. (2021). Racialized experiences differentiate food security among African American adults. *Nursing Research*, 70(5), S13-S20.
 8. *Ciesielski, T.H., Ngendahimana, D.K., Roche, A., Williams, S.M., Freedman, D.A. (2021). Elevated dietary inflammation among SNAP recipients provides targets for precision public health intervention. *American Journal of Preventive Medicine*.
 9. *Glickman A., Clark J.K., & Freedman D. (2021). A relational approach to the food environment finds that the food environment matters for those who use it. *Health & Place*.
 10. Jablonski, B.B.R., Casnovsky, J., Clark, J.K., Cleary, R., Feingold, B., Freedman, D., Gray, S., Romeiko, X., Olabisi, L.S., Torres, M., van den Berg, A.E., Walsh, C., & Wentworth, C. (2020). Emergency food provision for children and families during the COVID-19 pandemic: Examples from five U.S. cities. *Applied Economic Perspectives and Policy*.
 11. Sanchez-Flack, J., Joshi, K., Lee, E.E., & Freedman, D.A. (2021). Indicators of Readiness and Capacity for Implementation of Healthy Food Retail Interventions. *Journal of Agriculture, Food Systems, and Community Development*.
 12. Freedman, D.A., Clark, J.K., Lounsbury, D.W., Boswell, L., Burns, M., Jackson, M.B., Mikelbank, K., Donley, G., Worley-Bell, L., Mitchell, J., Ciesielski, T.H., Embaye, M., Lee, E.K., Roche, A., Gill, I., & Yamoah, O. (2022). Food system dynamics structuring nutrition equity in racialized urban neighborhoods. *American Journal of Clinical Nutrition*.
 13. Lee, E. K., Donley, G., Ciesielski, T. H., Gill, I., Yamoah, O., Roche, A., Martinez, R., Freedman, D. A. (2022). Health Outcomes in redlined versus non-redlined neighborhoods: A systematic review and meta-analysis. *Social Science & Medicine*. 294, 114696.

Book Chapter

1. Glickman, A.R., Clark, J.K., Freedman, D.A. (2022). Participatory modeling of the food system: The case of community-based systems dynamics. In: C. Peters (Ed.). *Food Systems Modelling*, 257-283. Academic Press, Cambridge, MA.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Research from the foodNEST 2.0 study is shared via the website of the Mary Ann Swetland Center for Environmental Health at Case Western Reserve University.

<https://case.edu/swetland/dissemination>

We also share findings via the Swetland Center Twitter handle: @CWRUSwetland

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Not applicable.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

All information products created during the Modeling the Future of Food in Your Neighborhood grant are available in the public domain through the Swetland Center website and/or through organizational websites. We actively share findings through public convenings and report via social media.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

Yes

1. Longitudinal cohort survey data (2 years)
2. Qualitative interview data
3. Causal loop diagramming data
4. Food environmental audit data (2 years)

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Yes

All data are from the same geospatial areas in Cleveland and Columbus, Ohio. The data represent diverse sources including qualitative and quantitative information from different sources. The data combine to inform how we conceptualize the food system and its dynamics and will be used to validate the modeling process both in the short- and long-term. The survey data are longitudinal.

- 4.3. **Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.**

The foodNEST 2.0 study was conducted in Cleveland and Columbus Ohio (2018-2021). The study included two annual surveys and 24-hour dietary assessments among residents of low-income urban neighborhoods, with the goal of learning about local food system experiences and diets. The survey questions were generally consistent throughout the study, but each year of data are available separately. This should allow users to access, manage, and analyze the primary source data in its original form, while properly accounting for key study features that differed between the years. Due to requirements of the consent, the de-identified data are housed at CWRU in a secure server and dates have been shifted in a manner that prevents identifiability but retains relational information. Each year of study data are included in a separate folder that contains 4 files:



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- 1) A pdf file summary of the Survey Questions, Answer Coding, and Basic Statistics for that year
- 2) A .csv excel file with the raw survey data from that year
- 3) A copy of the .csv excel file that includes the exact text of the questions and answers
- 4) A .csv excel file with the 24-hour nutritional intake data

Additional details about the study design and its implementation have been previously published:

Freedman DA, Bell BA, Clark J, et al. Small Improvements in an Urban Food Environment Resulted in No Changes in Diet Among Residents. J Community Health. March 2020:1-12. doi:10.1007/s10900-020-00805-z

<https://pubmed.ncbi.nlm.nih.gov/32170531/>

Ciesielski TH, Ngendahimana DK, Roche A, Elevated Dietary Inflammation Among Supplemental Nutrition Assistance Program Recipients Provides Targets for Precision Public Health Intervention Am J Prev Med. 2021 Aug;61(2):192-200. doi:

10.1016/j.amepre.2021.02.007 <https://pubmed.ncbi.nlm.nih.gov/33985837/>

Inquiries about the study and the data can be sent to:

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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____Darcy Freedman_____

PI Signature: __________ Date: _____6/1/2022_____

Authorized Signing Official (ASO) Name: _____Jon Donze_____

ASO Signature: _____ Date: _____